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MIL-B-20148D
AMENDMENT 1
14 July 1991

MILITARY SPECIFICATION

BRAZING SHEET, ALUMINUM ALLOY

This amendment forms a part of MIL-B-20148D, dated 30 March 1987, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2: Add under "Alloy":

"#21	Brazing Sheet (6951 Core, 4343 Clad) (formerly class 3)	Tempers O, F
#22	Brazing Sheet (6951 Core, 4343 Clad) (formerly class 4)".	Tempers O, F

Add at the end of the paragraph:
"(Information pertaining to classes #21 and #22 is contained in Appendix A - see 6.3.1)."

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2.1: Delete: "FED-STD-151 Metals; Test Methods".
Delete: "MIL-STD-649 Aluminum and Magnesium Products; Preparation for Shipment and Storage".

2.2: Add under "American Society for Testing and Materials":

"ASTM B 660	Standard Practice for Packaging/Packing of Aluminum and Magnesium Products
ASTM E 34	Standard Test Methods for Chemical Analysis of Aluminum and Aluminum Base Alloy".

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4.4.1: Delete "Method 111 of FED-STD-151" and substitute "ASTM E 34".

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5.1: Delete "MIL-STD-649" and substitute "ASTM B 660".

6.2.1: Delete "Numbers 11 and 12 ..." and substitute "Numbers 11, 12, 21, and 22 ...".

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6.2.2: Delete "Numbers 23 and 24..." and substitute "Numbers 21, 22, 23, 24...".

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6.3.1: Delete:

"However, upon verification that there are domestic producers manufacturing these materials, they will be reinserted in the next revision or amendment to this specification."

and substitute:

"The information pertaining to classes 3 and 4 of Revision C has been placed in Appendix A. When classes 3 and 4 (now 21 and 22) are required, the information in Appendix A, or portions thereof, shall be considered additional requirements when so directed by the Contracting Officer."

Add Appendix as follows:

"APPENDIX A

CLASS 21 AND 22 BRAZING SHEET

10. SCOPE

10.1 This appendix provides the information pertaining to classes 21 and 22 (formerly classes 3 and 4 respectively) brazing sheet. This information was deleted from revision D since no domestic producers of this type of brazing sheet could be found. The information supplied in this Appendix is for guidance and informational purposes only, unless otherwise specified in the contract or order.

10.2 Classes.

Class 21 - 6951 alloy clad one side with 4343 alloy
Class 22 - 6951 alloy clad both sides with 4343 alloy

20. DOCUMENTS
(This section is not applicable to this appendix.)

30. REQUIREMENTS

30.1 Tables. The tables listed herein contain information pertaining to the chemical composition (Table I), mechanical properties (Table II), and cladding thickness (Table III) requirements for classes 21 and 22 brazing sheet.

30.2 Tempers. When class 21 or class 22 brazing sheet is required in a definite temper other than those listed in Table II, the mechanical properties shall be specified in the contract or order.

30.3 Location of test specimens. Tension test specimens shall be taken from the sheet perpendicular to the direction of finish rolling for classes 21 and 22.

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TABLE I. Chemical composition. 1/ 2/

Elements	Core	Cladding
Silicon	0.20 - 0.50	6.8 - 8.2
Iron	0.8	0.8
Copper	0.15 - 0.40	0.25
Manganese	0.10	0.10
Magnesium	0.40 - 0.8	-----
Zinc	0.20	0.20
Other elements		
Each	0.05	0.05
Total 3/	0.15	0.15
Aluminum	Remainder	Remainder

1/ Analysis shall routinely be made only for the elements specifically mentioned in Table I. If, however, the presence of other elements is indicated or suspected in amounts greater than the specified limits, further analysis shall be made to determine that these elements are not present in excess of specified limits.

2/ Composition is maximum weight percent unless shown as a range.

3/ The sum of those "Other elements" 0.010 percent or more each, expressed to the second decimal before determining the sum.

TABLE II. Mechanical properties.

Class	Temper	Thickness (inches)	Tensile strength		Elongation In 2 Inches 1/ Minimum (percent)
			Minimum ksi	Maximum ksi	
#21 and #22	O	0.020-0.031	---	21	18
		0.032-0.050	---	21	20
		0.051-0.249	---	21	23
	F	ALL	2/	2/	2/

1/ Not required for material 0.5 inch or less in width.

2/ No mechanical test requirements.

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TABLE III. Cladding thickness.

Class	Total Thickness of Composite Product (inches) 1/	Number of Sides Clad	Thickness of Each Cladding Plate, nominal (% of total thickness)	Minimum Mean Thickness of Cladding on Finished Sheets (% of total thickness)	Maximum Mean Thickness of Cladding on Finished Sheets (% of total thickness)
21	$T \leq 0.090$	One	10	8	12
	$T \geq 0.091$	One	5	4	6
22	$T \leq 0.090$	Both	10	8	12
	$T \geq 0.091$	Both	5	4	6

1/ T = Total Thickness of Composite Material "

Custodians:
Army - MR
Air Force - 11
Navy - AS

Review Activities:
Army - ME, AR

Preparing Activity:
Navy - AS
(Project No. 3439-0788)

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